

# Adham Tarek Aladwash

Mechatronics Engineer | Junior Data Scientist & AI Developer

Damietta, Egypt • adhamaladwash71@gmail.com • +20 106 822 8935 • linkedin.com/in/adham-aladwash-8b3b8824a • github.com/ADHAMALADWASH

## OBJECTIVE

Undergraduate Mechatronics Engineering student and junior data scientist seeking to apply engineering and machine learning knowledge to real-world, AI-driven projects. Hands-on experience across automotive electrical systems, hybrid/EV diagnostics, industrial automation, and end-to-end data science pipelines.

## EDUCATION

**B.Sc. Mechatronics Engineering, Delta University for Science and Technology** Oct 2021 – Jun 2027  
GPA: 3.85 / 4.0 — Excellent Standing

## WORK EXPERIENCE

**Summer Intern — GB Academy** Jun 2026 – Jul 2026  
*Cairo, Egypt · On-site — Hybrid & Electric Vehicle Systems*

- Supported diagnostic and inspection procedures on hybrid/EV systems, including high-voltage battery packs, Battery Management Systems (BMS), and EV powertrain components.
- Performed electrical fault diagnosis using professional automotive diagnostic tools and OBD systems across multiple vehicle platforms.
- Inspected EV charging system architecture, including AC/DC charging protocols and onboard charging units.
- Conducted wiring harness inspection, connector integrity checks, and circuit continuity testing alongside senior engineers.
- Built working knowledge of regenerative braking systems, electric motor operation, and inverter functionality in hybrid/EV drivetrains.

**Intern — Data Scientist & AI Developer — Digital Egypt Pioneers Initiative (DEPI)** Jun 2025 – Jan 2026  
*Egypt · Remote · Part-time*

- Completed the IBM Data Scientist Track, covering the full data science pipeline from data preprocessing to model deployment.
- Co-developed SalesScope, a 6-month capstone project analyzing 83,000+ rows of sales data with machine learning forecasting, an interactive dashboard, and a support chatbot.
- Applied Python and SQL for data logging, cleaning, statistical analysis, and relational database design.

**Intern — Mechatronic Engineer — GB Academy** Jul 2025 – Sep 2025  
*Cairo, Egypt · On-site — Automotive Electrical Systems*

- Diagnosed and troubleshooted electrical and electronic faults in vehicle systems.
- Assisted with mechanical engine repairs and battery handling/recycling procedures.

## KEY PROJECTS

**Automated Pneumatic Garbage Sorting System** Mar – May 2026

- Designed and built a zero-intervention recyclable sorting system using an inductive proximity sensor and TCS3200 color sensor feeding an Arduino-controlled pneumatic system.
- Achieved 3× faster sorting than manual labor and an estimated 40% reduction in operating cost, with millisecond-level response time.

**3-DOF Robotic Arm — ESP32 Wireless Control** Mar – May 2026

- Built a 3-DOF robotic arm with real-time control via a custom web app and voice commands, using an ESP32 WebSocket server and Wi-Fi Access Point mode requiring no internet connection.

**Real-Time Smart City Monitoring Ecosystem** Feb – May 2026

- Developed an end-to-end IoT monitoring system streaming ESP32 sensor data to a Python/Flask backend, applying machine learning (Isolation Forest, classification, regression) for real-time anomaly detection and forecasting.

### **SalesScope — End-to-End Sales Analytics & Forecasting System**

*Jun – Dec 2025*

- Built a deployed pipeline combining data preprocessing, ML forecasting models, an interactive dashboard, and a support chatbot on a dataset of 83,000+ sales records.

## **CERTIFICATIONS & TRAINING**

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- Practical Training — Classic Control, Electric Motor & Drive Programming, PLC Programming, HA Consulting Group (Jun–Aug 2026, Grade: Excellent) — Siemens, Schneider Electric, WECON, and Kinco PLCs; Yaskawa J1000 and Delta VFDs; CAdE SIMU and Factory IO simulation.
- CS50: Introduction to Computer Science, Harvard University / edX (Aug–Oct 2022).
- C Programming: Language Foundations, Dartmouth College / edX (Aug–Sep 2023).
- Embedded C: Hardware Essentials and Device Driver Development.
- Motorsport Engineer – Introductory Course.
- Car Repair and Electrician Training Certificate (CRETC+).
- Data Analysis, Deep Learning & Soft Skills Program, MCIT.

## **SKILLS**

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**Programming:** Python, C, C++, SQL, Embedded C

**Data Science & AI:** Machine Learning, Deep Learning, Data Analysis, Pandas, NumPy, Scikit-Learn

**CAD & Mechanical:** SolidWorks, AutoCAD, 3D Modeling, 3D Printing, FluidSIM

**Industrial Automation:** PLC Programming, Ladder Logic, Siemens TIA Portal, Schneider Electric, WECON, Kinco, Factory IO

**Automotive & EV:** Electric Vehicles, Battery Management Systems, EV Powertrain, OBD Diagnostics, AC/DC Charging, Inverter Systems

**Interpersonal:** Time Management, Team Leading, Negotiation